

E.3 / PAPER 2 / QUESTION 7

26 April 2024 / TZ1 (file label) / HL Paper 2 / original Q6 / 2224-6702

7. (a) State what is meant by the half-life of a radioactive nuclide.

[1]

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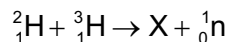
Tritium, ${}^3_1\text{H}$, is a radioactive isotope of hydrogen. The activity of a sample of tritium decreases to $\frac{A_0}{8}$ after a time of 37.0 years where A_0 is the initial activity.

- (b) Calculate the half-life of tritium.

[2]

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Deuterium, ${}^2_1\text{H}$, and tritium undergo nuclear fusion according to:



- (c) Identify nuclide X.

[1]

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(Question 7 continued)

The following data are given for binding energies per nucleon.

Nuclide	Binding energy per nucleon / MeV
${}^2_1\text{H}$	1.112
${}^3_1\text{H}$	2.827
X	7.074

- (d) (i) Show that the energy released in this reaction is about 18 MeV. [2]

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- (ii) Hence, estimate the energy, in J, released as a result of fusion of 1 kg of nuclear fuel according to this reaction. [2]

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- (iii) The specific energy of typical fossil fuels is about 10^7 J kg^{-1} .

Outline, with reference to your answer in (d)(ii), the advantage of nuclear fusion over fossil fuels as a source of energy. [1]

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